Computer Architecture – MCQ and Numerical Questions (with Answers)

Based on: Computer Architecture: A Quantitative Approach (Hennessy & Patterson, 6th Ed.)

Section A: Multiple Choice Questions (MCQs)

1. Which metric best represents CPU performance?
A) Clock rate
B) CPI
C) Execution time
D) Instruction count
Answer: C

2. According to Amdahl's Law, system speedup is mainly limited by:
A) Enhanced portion
B) Unenhanced portion
C) Clock speed
D) Memory size
Answer: B

3. Which memory has the fastest access time?
A) DRAM
B) Flash
C) SRAM
D) HDD
Answer: C

4. Cache miss penalty refers to:
A) Time to check cache
B) Time to fetch block from next level
C) CPU stall time only
D) Cache write time
Answer: B

5. Which cache mapping gives lowest conflict misses?
A) Direct mapped
B) Fully associative
C) Set associative
D) Random mapped
Answer: B

6. CPI stands for:
A) Cycles per instruction
B) Clocks per interrupt
C) Cost per instruction
D) Cycles per interrupt
Answer: A

7. Pipeline hazard caused by branch instructions is called:
A) Structural hazard
B) Data hazard
C) Control hazard
D) Memory hazard
Answer: C

8. SIMD is mainly useful for:
A) Control flow
B) Scalar operations

C) Data parallel operations
D) I/O operations
Answer: C

9. Which technique allows multiple instructions per cycle dynamically?
A) Pipelining
B) Superscalar
C) Multithreading
D) Vector processing
Answer: B

10. GPU architectures are optimized for:
A) Low latency
B) High single-thread performance
C) High throughput
D) Large caches
Answer: C

11. Directory-based cache coherence is mainly used in:
A) Single-core CPUs
B) SMP systems only
C) Large-scale multiprocessors
D) GPUs
Answer: C

12. Which memory consistency model is the strongest?
A) Weak consistency
B) Release consistency
C) Sequential consistency
D) Relaxed consistency
Answer: C

13. MapReduce is mainly used in:
A) Embedded systems
B) Warehouse-scale computers
C) GPUs
D) Personal computers
Answer: B

14. Power = $C \times V^2 \times f$ refers to:
A) Static power
B) Dynamic power
C) Leakage power
D) Thermal power
Answer: B

15. Which factor mainly limits frequency scaling today?
A) Transistor size
B) Memory speed
C) Power density
D) Disk latency
Answer: C

16. Which ISA is used in the 6th edition examples?
A) MIPS
B) ARM
C) x86
D) RISC-V
Answer: D

17. Vector length register helps in:
A) Branch prediction

B) Handling variable vector sizes
C) Cache replacement
D) Address translation
Answer: B

18. Which improves ILP by predicting branch direction?
A) Scoreboarding
B) Tomasulo's algorithm
C) Branch predictor
D) Register renaming
Answer: C

19. Which memory is non-volatile?
A) SRAM
B) DRAM
C) Cache
D) Flash
Answer: D

20. Misses per instruction =
A) Miss rate × CPI
B) Miss rate × Memory accesses per instruction
C) Hit rate × CPI
D) Miss penalty × CPI
Answer: B

21. Throughput-oriented processors focus on:
A) Single-thread latency
B) Power reduction only
C) Total work per unit time
D) Disk performance
Answer: C

22. Which supports multiple hardware threads per core?
A) SIMD
B) SMT
C) Vector
D) GPU warp
Answer: B

23. Roofline model relates performance to:
A) Clock speed
B) Memory bandwidth and operational intensity
C) Cache size
D) Number of cores
Answer: B

24. Which is an example of domain-specific architecture?
A) Intel Core i7
B) ARM Cortex A53
C) Google TPU
D) DDR4 DRAM
Answer: C

25. Availability =
A) MTTF × MTTR
B) MTTR / MTTF
C) MTTF / (MTTF + MTTR)
D) MTTR / (MTTF + MTTR)
Answer: C

Section B: Numerical Problems

1. A program executes 1×10■ instructions with CPI = 1.5 and clock rate = 2 GHz. Find CPU time.
Answer: Time = (IC × CPI) / Clock rate = (1e9 × 1.5) / 2e9 = 0.75 s

2. If 40% of execution time can be improved by a factor of 5, find overall speedup.
Answer: Speedup = 1 / (0.6 + 0.4/5) = 1 / (0.6 + 0.08) = 1 / 0.68 ≈ 1.47

3. Cache hit time = 1 ns, miss rate = 5%, miss penalty = 50 ns. Find AMAT.
Answer: AMAT = 1 + 0.05 × 50 = 3.5 ns

4. Miss rate = 2%, memory accesses per instruction = 1.2. Find misses per instruction.
Answer: MPI = 0.02 × 1.2 = 0.024

5. If voltage is reduced by 10% and frequency by 10%, dynamic power changes by:
Answer: $P \propto V^2f \rightarrow (0.9^2 \times 0.9) = 0.729 \rightarrow$ 27.1% reduction

6. Two systems run same workload. A takes 20s, B takes 12s. How many times faster is B?
Answer: Speedup = 20 / 12 = 1.67×

7. A cache has 32 KB size, block size 64 B, 4-way set associative. Find number of sets.
Answer: Sets = 32KB / (64 × 4) = 32768 / 256 = 128 sets

8. If MTTF = 1000 hours and MTTR = 10 hours, find availability.
Answer: Availability = 1000 / (1010) ≈ 0.9901 (99.01%)

9. A GPU sustains 2 TFLOPS and memory bandwidth is 250 GB/s. Find operational intensity at balance point.
Answer: OI = Performance / Bandwidth = 2000 / 250 = 8 FLOP/byte

10. If CPI without cache misses is 1.0 and miss CPI is 0.4, find total CPI.
Answer: Total CPI = 1.0 + 0.4 = 1.4